

Howard Physics Colloquium, How AI and Physicists Think – James Yorke, September 9, 2026 Lessons for Study

In “How AI and Physicists Think”, mathematician James A. Yorke examines how people learn, develop intuition, and solve difficult problems - and what artificial intelligence may teach us about these processes.

Yorke distinguishes three interacting forms of thought:

1. **Type 1: intuition** — ideas, patterns, and possible solutions that arise rapidly and often subconsciously.
2. **Type 2: deliberate reasoning** — consciously examining those ideas, following logical steps, testing alternatives, and constructing an argument.
3. **Type 3: training** — the sustained practice through which knowledge and experience gradually improve both intuition and reasoning.

His central claim is that sophisticated reasoning is often a sequence of intuitive judgments. A scientist first recognizes a promising possibility, then investigates its consequences through careful reasoning. Using chess as an analogy, Yorke argues that fast and slow play depend on many of the same learned intuitions: extended thought allows a player to explore a larger tree of possibilities, but each branch is still evaluated through trained judgment.

Yorke illustrates this process through Einstein’s work. Einstein’s breakthroughs began with simple but radical questions: Why should moving a magnet and moving a coil require different descriptions? What if acceleration and gravity are physically indistinguishable? What if light itself comes in discrete packets? The mathematics required enormous effort, but the decisive starting points were imaginative conceptual insights.

The lecture therefore challenges conventional education. Students often spend many years answering questions selected by instructors, yet research depends heavily on learning to formulate questions of their own. Yorke argues that education should allow students to struggle, make mistakes, revisit material, discover relationships, and develop intuition-not merely memorize procedures or advance rapidly through a curriculum.

AI adds urgency to this issue. When answers and explanations are readily available, education cannot be limited to retrieving information or reproducing standard solutions. Students must learn how to ask productive questions, evaluate answers, connect ideas, and continue learning independently. Yorke presents conversational AI as a potentially powerful personal tutor: it can help someone enter an unfamiliar field by answering successive questions at the learner’s level. But easy access to answers also creates the danger that students will stop doing the difficult intellectual work required to build their own intuition.

The lecture’s larger message is that expertise does not consist simply of knowing more facts. It arises from prolonged engagement with problems-trying ideas, making mistakes, recognizing patterns, and learning which possibilities deserve deeper investigation.

The argument in one sentence

The purpose of education should be to cultivate trained intuition, deliberate reasoning, independent learning, and the ability to ask worthwhile questions-because those capacities become more important, not less, when AI can provide immediate answers.

Particularly useful lessons for students

- Learn how to teach yourself; formal education cannot prepare you for everything you will eventually need to know.
- Ask questions instead of waiting only to receive them.
- Treat mistakes as part of the mechanism by which intuition is trained.
- Do not confuse rapid curricular advancement with deep understanding.
- Revisit important material: understanding may continue growing long after an examination.
- Use AI to interrogate unfamiliar ideas, not merely to avoid thinking.
- Remember that a brilliant insight still requires sustained work before it becomes defensible science.